Covid-19 Global Health Analytics

Chanh Minh Dat Luc

Table of Contents

1.1.1 Introduction 2
2. Analytics Solutions 3
  2.1.1 Excel Analytics 3
  2.1.2 How does pandemic burden compare across continents? 3
  2.1.3 How does healthcare readiness vary across countries in different continents? 7
  2.2.1 Power Bi Analytics 9
  2.2.2 Regional Vulnerability and Mortality Outcomes 9
  2.2.3 How did the pandemic evolve over time across continents 10
  2.2.4 How did recovery capacity differ across regions? 12
  2.2.5 Testing and mortality disparities across continents 13
3.1.1 Recommendations 14
  3.1.2 Expand healthcare infrastructure in vulnerable regions 14
  3.1.3 Strengthen Vaccination access and equity 14
  3.1.3 Scale Up Testing and Surveillance Systems 15
  3.1.4 Invest in Community Resilience and Recovery Capacity 15
4.1.1 Conclusion 15
5.1.1References 16
6.1.1 Appendix 19
  6.1.2Power Bi Dashboard 19
  6.1.3 Excel Dashboard 20

1.1.1 Introduction

The COVID-19 pandemic has been one of the most profound global health crises in modern history, exposing weaknesses in healthcare systems and testing the resilience of societies worldwide. Beyond the immediate toll of widespread morbidity and mortality, the pandemic revealed stark disparities in preparedness and response capacity between countries. High-income nations generally benefited from stronger medical infrastructure, broader access to vaccines, and advanced surveillance systems, whereas low- and middle-income countries often struggled with limited healthcare resources, logistical bottlenecks, and inequitable vaccine distribution (Ayoade Alakija, 2023). These disparities created highly uneven outcomes and reinforced the urgent need for targeted international support.

As the global community moves into the post-pandemic era, the World Health Organization (WHO) faces critical decisions regarding the allocation of resources to improve pandemic preparedness, particularly in vulnerable countries with weak health systems (World Health Organization, 2022). Effective allocation requires robust, evidence-based insights into historical COVID-19 data, including trends in infections, deaths, vaccination uptake, and healthcare capacity. Such data-driven analysis not only helps to identify nations at heightened risk but also informs strategic planning for strengthening health systems and building resilience against future health emergencies (Ayogeboh Epizitone et al., 2023).

This report addresses these challenges by applying visual analytics tools: Microsoft Excel and Power BI to analyze historical COVID-19 data and derive actionable insights. Excel is used to construct an accessible, descriptive dashboard that highlights global and regional trends in cases, deaths, and vaccination uptake. Power BI complements this analysis by enabling interactive exploration of healthcare system capacity and country-level vulnerabilities, with features such as drill-through analysis and dynamic mapping. Together, these analytics solutions provide a multi-dimensional understanding of pandemic outcomes, illuminating patterns that might otherwise remain obscured in raw data.

The report is structured to support WHO leadership in evidence-based decision-making. First, the developed dashboards are presented and explained, with key insights extracted from both Excel and Power BI solutions. These insights are then translated into specific, actionable recommendations on how the WHO can most effectively allocate resources to strengthen public health infrastructure, expand vaccination coverage, and enhance surveillance capabilities in vulnerable regions. Importantly, the recommendations are grounded not only in the data analysis but also in relevant academic and policy literature, ensuring that proposed strategies align with broader global health priorities.

Additionally, the report acknowledges the limitations and gaps in the provided dataset. Variability in reporting standards, underreporting in low-income countries, and limited integration of socioeconomic indicators pose challenges to drawing definitive conclusions. Recognizing these limitations, the report also suggests ways to improve future data collection, including the incorporation of wider demographic, economic, and health system variables to enhance the quality and reliability of the analysis.

Ultimately, the purpose of this report is to bridge the gap between complex pandemic data and an actionable global health strategy. By leveraging visual analytics and evidence-based reasoning, it seeks to provide WHO with clear, data-supported guidance on resource allocation. Strengthening public health systems in the most vulnerable regions is not only a matter of equity but also a global imperative, as the next pandemic will test the preparedness of the world.

2. Analytics Solutions

2.1.1 Excel Analytics

Excel is one of the insight generators from Microsoft that helps Data Analysts at WHO like us to comprehensively determine, identify, and compare the relationships between data and data (Abdullah Al Maruf et al., 2022). In other words, this tool allows us to identify any bottlenecks or difficulties that Covid-19 has brought to our world and process urgent action as soon as possible for those countries with high vulnerability.

In this case, pivot tables and charts will be used to create insights that may elevate leaders and managers at WHO with better decision-making on what should do and process next steps during this pandemic.

2.1.2 How does pandemic burden compare across continents?

The Excel Dashboard compared the top 15 countries across continents by cases and deaths per one million population. These insights highlight the distinct differences in how the pandemic burden was distributed globally.

- Asia: Bahrain and the Maldives recorded very high infection rates (cases) relative to population. However, countries like Armenia and Georgia showed a higher ratio of deaths to cases, symbolizing weaker healthcare systems and slower responses.

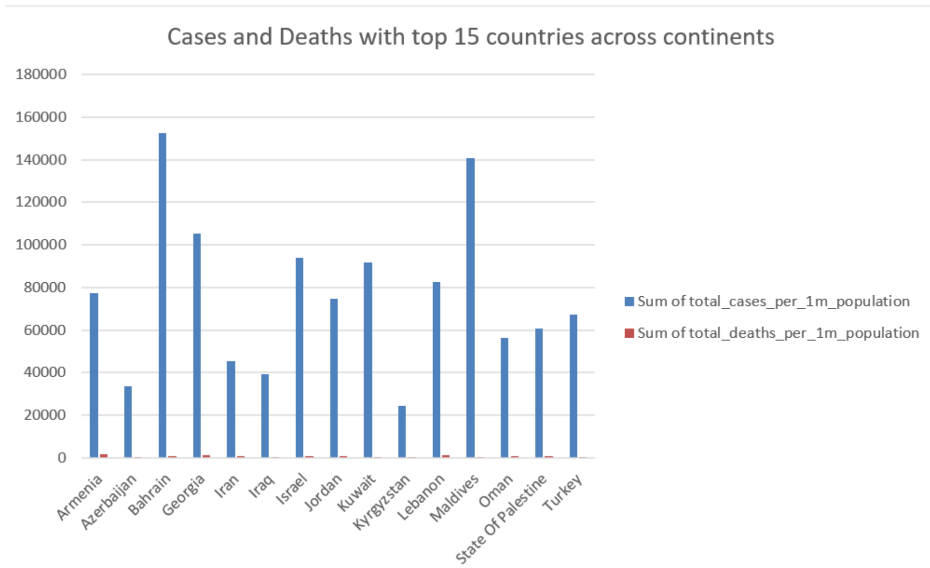


Figure 1: Cases and Deaths with the top 15 countries in Asia

- Australia/Oceania: French Polynesia and Wallis & Futuna have the highest case rates in the region. Deaths per 1m are concentrated in Wallis & Futuna and the Marshall Islands, with Fiji also elevated. Australia and New Zealand have very low deaths per 1 million.

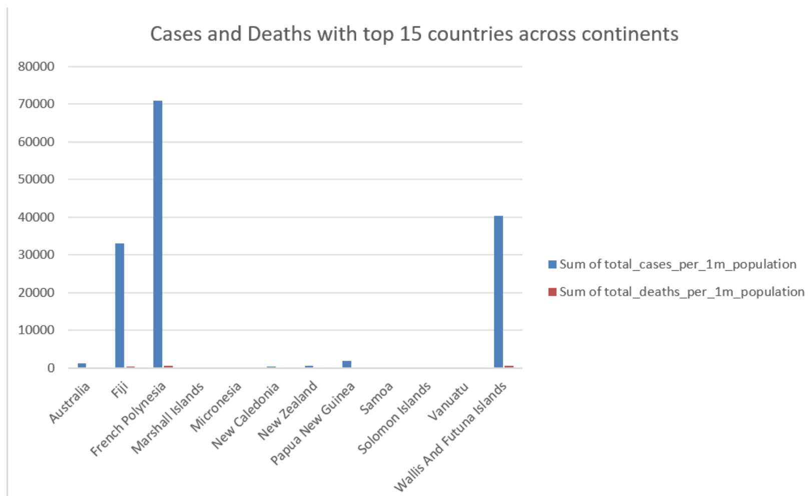


Figure 2: Cases and Deaths with the top 15 countries in Autralia/Oceania

- Europe: Very high cases per 1 million in Montenegro (~162k), the Czech Republic (~156k), San Marino (~151k), Gibraltar (~147k), and Slovenia (~125k). Deaths per 1 million are highest in Hungary (~3,117), Bosnia & Herzegovina (~2,973), the Czech Republic (~2,831), San Marino (~2,646), and Bulgaria (~2,642). The UK (~1,899) is lower than these peaks but still substantial.

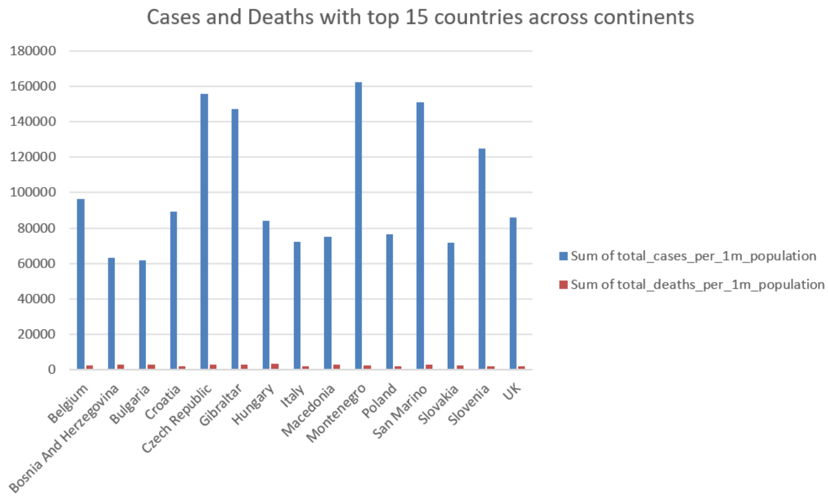


Figure 4: Cases and Deaths with the top 15 countries in Europe

- North America: Highest cases per 1 million in Aruba (~109k), USA (~107k), and Panama (~99k). Deaths per 1 million are highest in the USA (~1,889) and Mexico (~1,844), with Panama (~1,552) also high. Canada (~698) is clearly lower in deaths per 1 million.

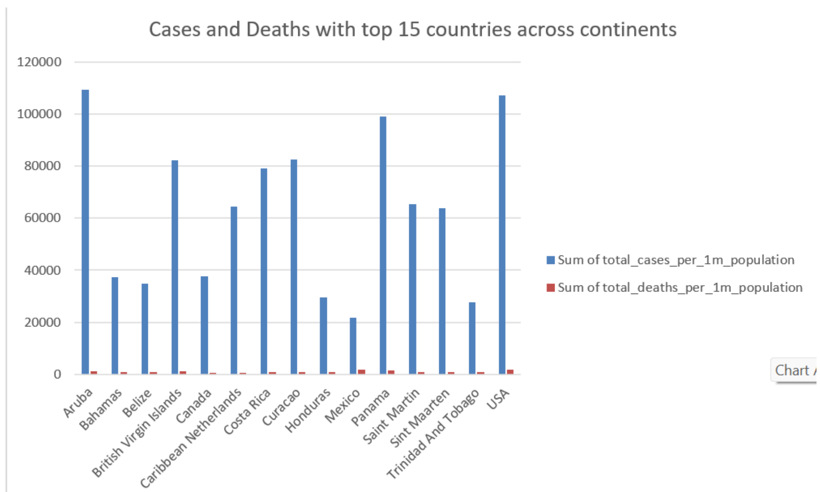


Figure 5: Cases and Deaths with the top 15 countries in North America

- South America: High cases per 1 million in Uruguay (~109k) and Argentina (~108k), with Brazil (~93k) and Colombia (~93k) close behind. Deaths per 1 million peak sharply in Peru (~5,865), far above Brazil (~2,598), Argentina (~2,316), Colombia (~2,346), Paraguay (~2,066), and Uruguay (~1,709).

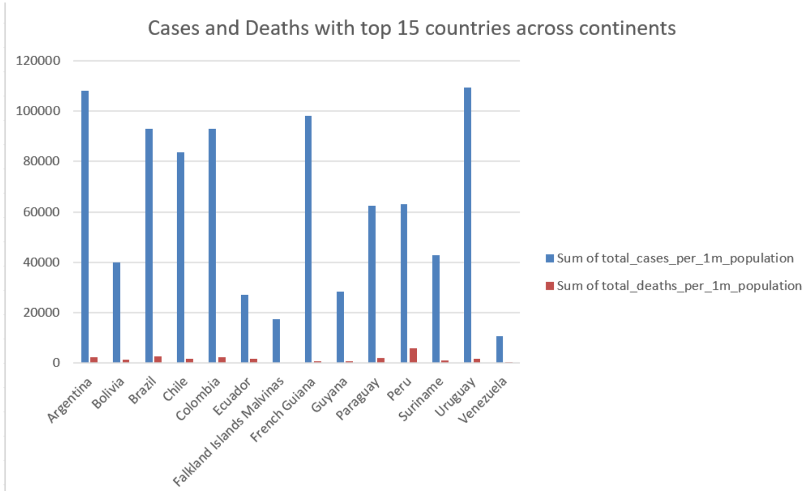


Figure 6: Cases and Deaths with the top 15 countries in South America

- Africa: Seychelles shows the highest cases per 1 million in the region (by a large margin on your chart). Deaths per 1 million are highest in Tunisia (~1,647), South Africa (~1,198), and Namibia (~1,175), followed by Seychelles (~949) and Cabo Verde (~530). Botswana (~653) and Libya (~503) are also elevated.

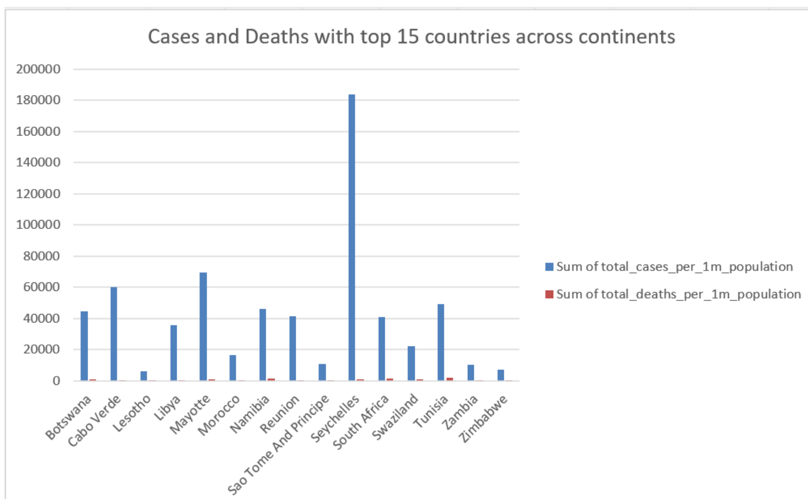


Figure 7: Cases and Deaths with the top 15 countries in Africa

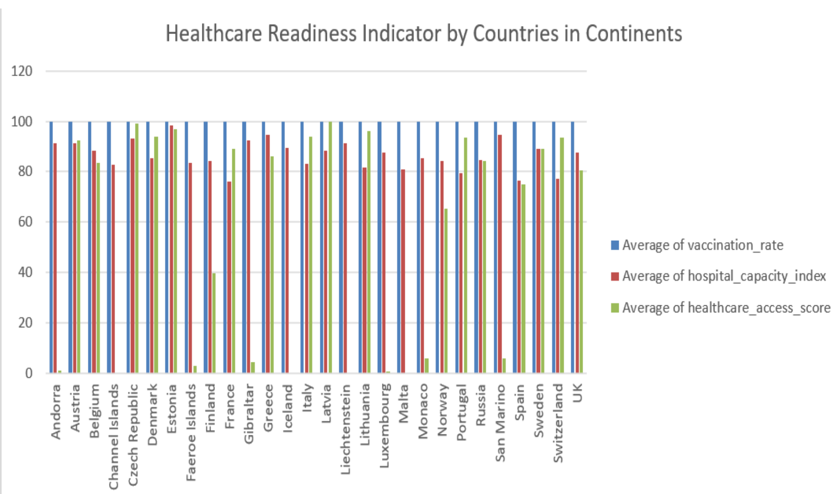
The analysis highlights that the burden of COVID-19 was unevenly distributed across continents. Europe and the Americas recorded the highest case rates per million, reflecting widespread outbreaks and stronger testing capacity, but still faced significant mortality pressures. South America combined high infection rates with some of the world's highest death rates, particularly in Peru, due to weaker healthcare infrastructure (Taylor, 2021). Asia showed mixed outcomes: countries such as Bahrain and the Maldives had very high case rates but lower deaths, while Armenia and Georgia reported higher death-to-case ratios, pointing to fragile health systems (Chowdhury et al., 2020)

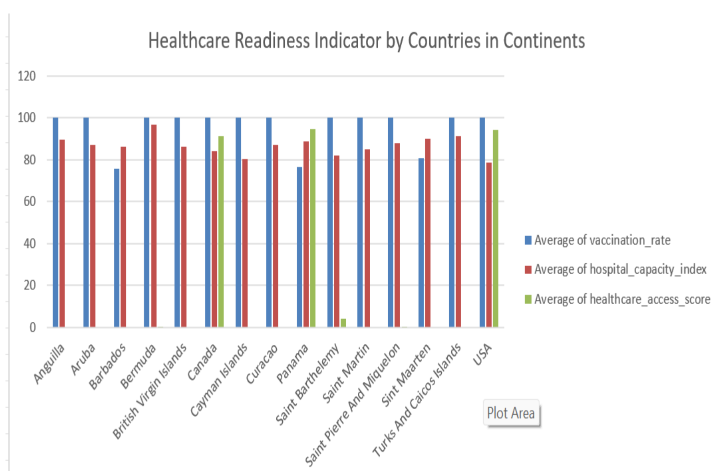
. Africa reported relatively low case numbers, but elevated death rates in Tunisia, Namibia, and South Africa suggest under-testing and limited healthcare capacity (Haider et al., 2020). In contrast, most of Australia and Oceania experienced lower burdens due to isolation and strict border policies, though smaller island nations such as Fiji and Wallis & Futuna struggled when outbreaks occurred. Overall, case numbers alone do not reflect true vulnerability; higher death ratios in regions with weak systems underline the need for WHO to prioritize support where healthcare capacity is most limited.

2.1.3 How does healthcare readiness vary across countries in different continents?

The Excel dashboard also compared three indicators of healthcare readiness: vaccination rate, hospital capacity index, and healthcare access score. The results highlight strong inequalities between regions. Europe and North America recorded the highest vaccination coverage (close to or at 100% in many countries), with consistently strong hospital capacity and access scores. Asia showed a mixed picture: wealthy states such as Bahrain, Israel, and Singapore achieved high vaccination and hospital capacity levels, while other countries remained lower. South America averaged around 46% vaccination, with countries like Chile and Uruguay achieving high coverage, but others, such as Bolivia and Venezuela, lagging far behind. Australia and Oceania displayed stark contrasts: Australia reached above 95% vaccination, but many Pacific Island nations had very low rates, in some cases below 10%. Africa showed the lowest overall readiness, with average vaccination below 30% and healthcare access scores remaining weak despite reasonable hospital capacity indices.

Insight: Countries with high vaccination and strong access scores (Europe, North America, parts of Asia) were better positioned to manage pandemic waves, while regions with low coverage and weaker access (Africa, parts of Oceania, and South America) remained highly vulnerable. This underlines the need for WHO resource allocation to focus on expanding





vaccination programs and strengthening healthcare access in low-income regions(World Health Organization, 2021).

Figure 8: Healthcare Readiness Indicator by top 15 countries in Europe (left) and North America (right)

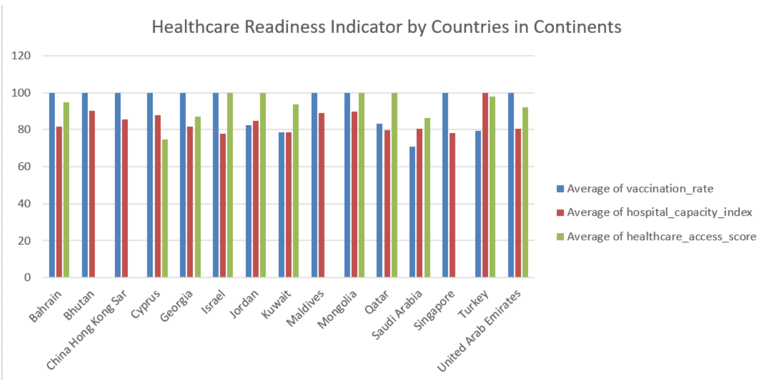


Figure 9: Healthcare Readiness Indicator by top 15 countries in Asia

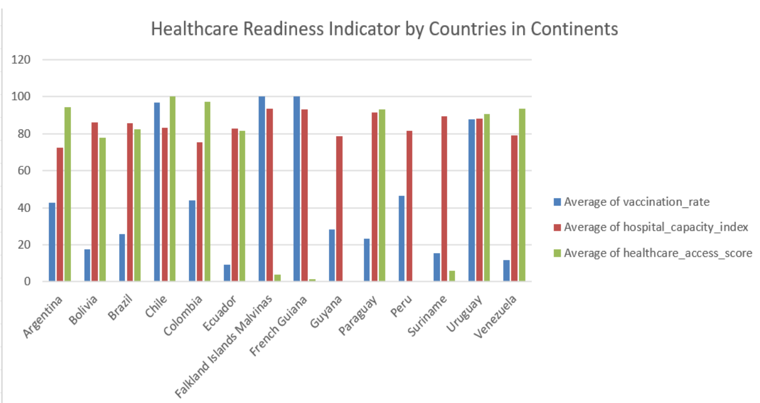


Figure 10: Healthcare Readiness Indicator by top 15 countries in South America

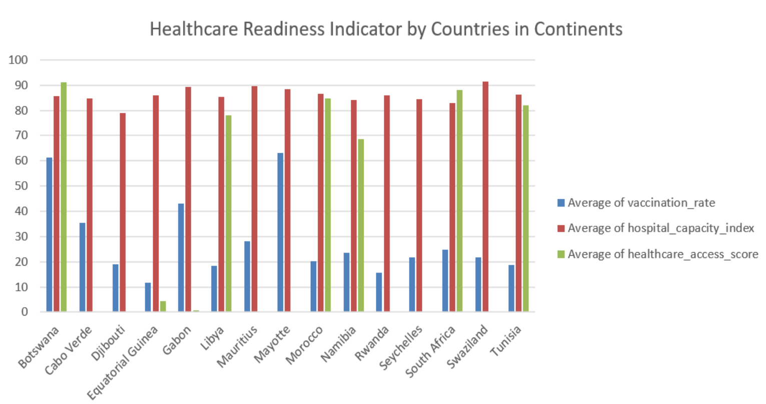


Figure 11: Healthcare Readiness Indicator by top 15 countries in Africa

2.2.1 Power Bi Analytics

Power BI was used to create the second dashboard, focusing on interactive and diagnostic analysis of the COVID-19 pandemic. Unlike the Excel dashboard, which summarized descriptive trends, Power BI enabled dynamic filtering, drill-through comparisons, and visualisation of complex relationships across time and regions. This broader, interactive perspective allowed for a deeper understanding of vulnerabilities and response capacity worldwide.

The dashboard incorporated several advanced features:

- **Time-series trends** to track the evolution of outbreaks and identify the timing and severity of pandemic waves.
- **Heatmaps and bubble charts** show the relative scale of infections and deaths across regions, highlighting countries with extreme burdens.
- **Recovery rates and testing levels** to assess which countries managed to maintain control and which fell behind.

These insights go beyond reporting case and death totals. They reveal structural weaknesses in healthcare systems, highlight the role of vaccination and testing in shaping outcomes, and provide WHO decision-makers with evidence to prioritize support in regions with limited capacity to respond effectively.

2.2.2 Regional Vulnerability and Mortality Outcomes

One of the key visuals in the Power BI dashboard is the global heatmap of deaths per million population. Each bubble represents a country, with the size of the bubble indicating the severity of mortality relative to population size. This view provides an immediate picture of where the pandemic was most deadly and which regions showed higher vulnerability.

The heatmap highlights several important insights:

- **Europe and South America** display clusters of large bubbles, confirming that these regions experienced some of the highest death rates relative to their populations. Countries such as Hungary, Bosnia, and Peru were among the most severely affected.
- **North America**, particularly the United States and Mexico, also shows very large bubbles, reflecting both high case volumes and elevated mortality.

- **Africa** reveals a more scattered pattern. While reported cases were lower, certain countries, such as South Africa and Tunisia, display larger bubbles, suggesting high mortality relative to case detection and weaker healthcare capacity.
- **Asia and Oceania** show more variation. Countries such as Armenia and Georgia recorded significant mortality burdens, while others, including Australia and several Pacific Island nations, reported relatively smaller bubbles due to stronger public health measures or smaller outbreaks.

Key insight: The heatmap shows that vulnerability was not evenly spread. Regions with limited health infrastructure and delayed vaccination campaigns, particularly in South America and parts of Africa, experienced disproportionately high mortality (Santangelo et al., 2024). For WHO, this highlights the need to focus future resource allocation not only on absolute case counts but also on strengthening system resilience in countries where death-to-case ratios reveal hidden vulnerability (World Health Organisation, 2017).

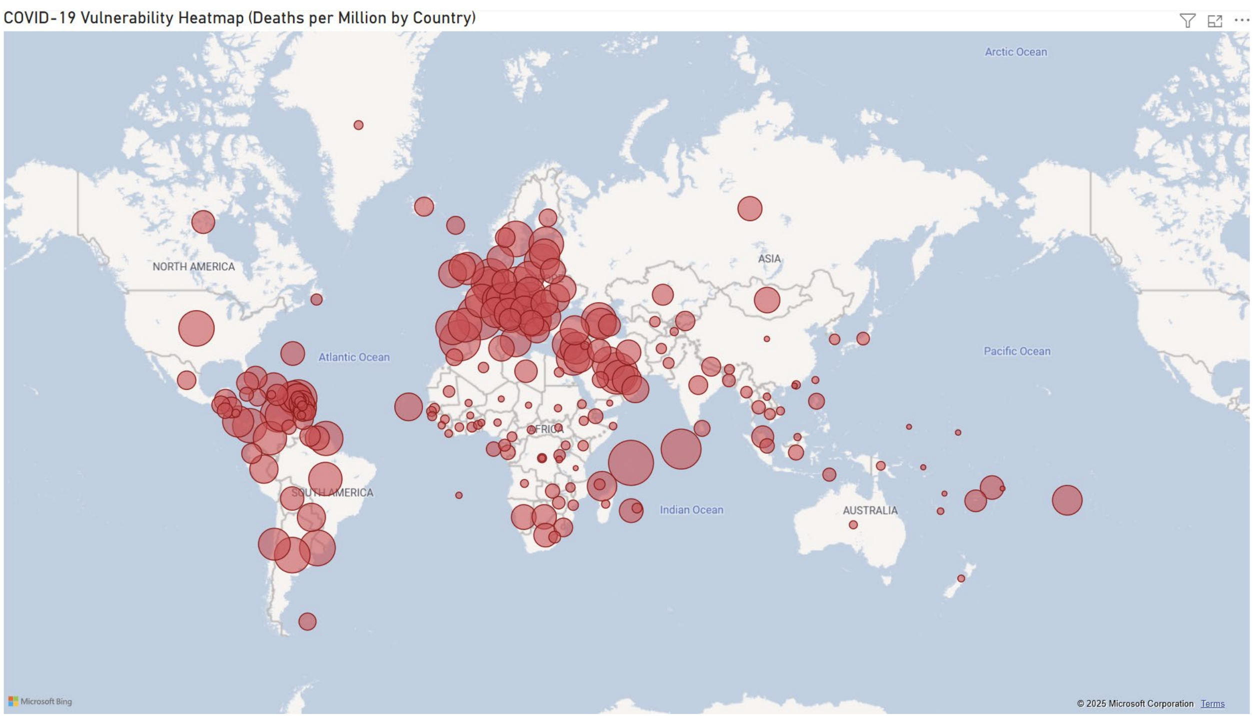


Figure 12: Covid-19 Vulnerability Heatmap

2.2.3 How did the pandemic evolve over time across continents

The Power BI time-series visual tracks daily new cases and daily new deaths globally from 2020 to mid-2021. This interactive feature allows filtering by continent to see how each region experienced the pandemic waves.

The timeline reveals three clear phases:

- **Early 2020:** The first global outbreak saw steep rises in both cases and deaths, particularly in Europe and North America.
- **Late 2020 to early 2021:** The second wave marked the sharpest peak, with cases surpassing 7 million and deaths reaching their highest levels worldwide. This coincided with the emergence of more transmissible variants and occurred before vaccines were widely rolled out.
- **Mid-2021:** A marked decline in deaths followed mass vaccination campaigns in Europe and North America, even though case numbers remained high. Regions with lower vaccination coverage, including Africa and parts of South America, continued to show slower declines in mortality.

Key insight: The time-series confirms that pandemic severity was shaped not only by infection waves but also by the timing of interventions such as vaccine distribution. Countries with earlier vaccination rollouts saw deaths fall more sharply after 2021, while others without widespread access remained vulnerable. This highlights the importance of global equity in vaccine distribution for future outbreak control (Wu et al., 2025).

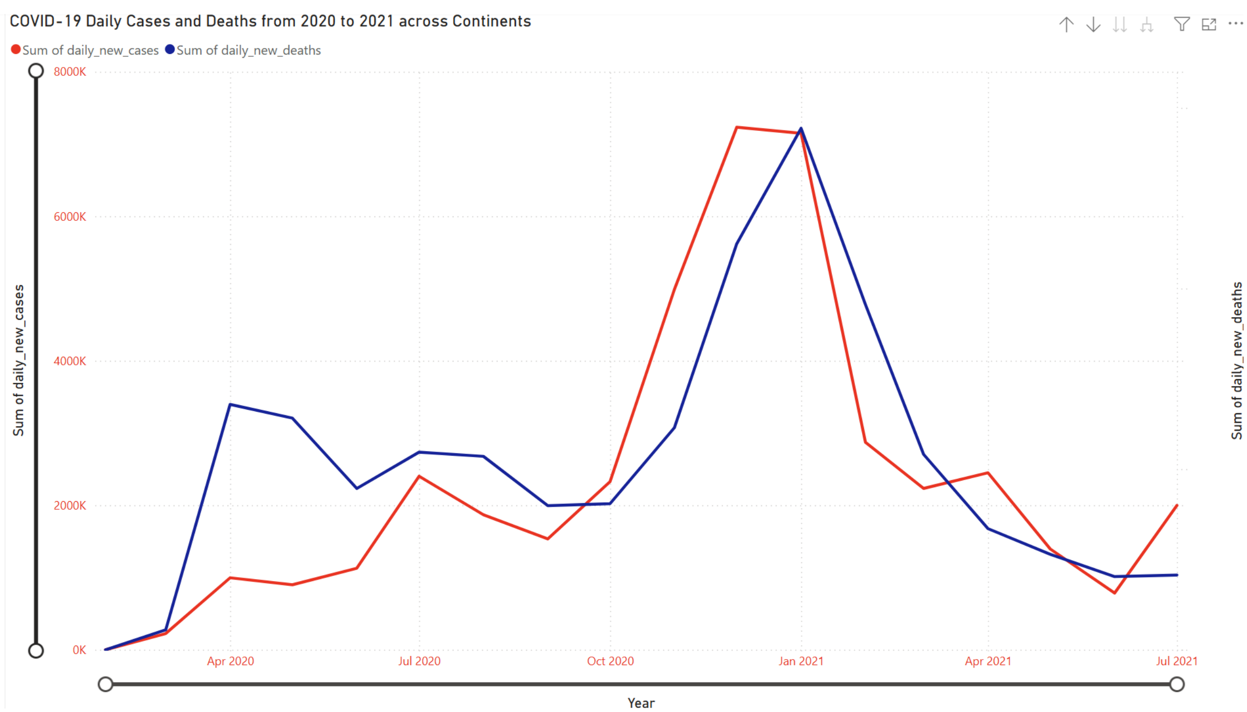


Figure 13: COVID-19 Daily Cases and Deaths from 2020 to 2021 across Continents

2.2.4 How did recovery capacity differ across regions?

The Power BI dashboard also compared total confirmed cases, deaths, active cases, and recovery rates by continent. This provides an important measure of how effectively different regions managed the pandemic once cases were detected.

- **Africa:** With about 7 million confirmed cases and 170,000 deaths, Africa recorded a recovery rate of roughly **85%**. While this is relatively strong, under-testing may mask the true scale of infections and limit the reliability of recovery estimates.

- **Asia:** The continent reported more than 62 million confirmed cases and nearly 900,000 deaths, with a recovery rate close to **88%**. Stronger health infrastructure in parts of Asia helped reduce deaths, although outcomes varied widely across countries.

- **Australia/Oceania:** With only 105,000 confirmed cases and 1,537 deaths, recovery data was incomplete in the dashboard. However, the region overall showed high resilience due to strict border controls and smaller outbreak sizes.

- **Europe:** Europe experienced over 51 million confirmed cases and 1 million deaths, yet achieved a recovery rate above **92%**, the highest among continents. This reflects both strong healthcare capacity and widespread vaccine access.

- **North America:** With more than 43 million confirmed cases and 940,000 deaths, recovery rates averaged around **82%**, lower than Europe and Asia. High mortality in countries such as the United States and Mexico contributed to this outcome.

- **South America:** The region recorded 36 million confirmed cases and about 1 million deaths, with a recovery rate of **93%**. Despite high infection levels, improved treatment capacity and vaccine uptake in some countries contributed to higher recovery rates compared with North America.

Key insight: Recovery rates demonstrate that stronger healthcare systems and vaccination access reduced mortality and improved outcomes (Skolnik et al., 2021). Europe and South America showed the highest recovery rates, while North America and parts of Africa lagged. For WHO, these disparities highlight the importance of strengthening recovery capacity in regions where access to care and treatments remains limited.

2.2.5 Testing and mortality disparities across continents

The Power BI dashboard also visualized the distribution of total deaths and total tests across continents. Comparing these two measures highlights important gaps in detection and response.

The death distribution shows that Europe (26.8%), South America (25.8%), and North America (22.2%) accounted for the largest shares of global mortality. Asia contributed around 21%, while Africa and Oceania reported only a small proportion of total deaths.

Testing data, however, reveals a different pattern. Asia (36.6%) and Europe (35.9%) conducted the highest number of tests, followed by North America (19.9%). By contrast, South America (4.8%), Africa (~2%), and Oceania (~1%) conducted far fewer tests.

Key insight: The comparison shows that continents with high testing (Europe, Asia, North America) detected more cases and deaths but also maintained better surveillance and outbreak control. In contrast, Africa and South America reported relatively fewer tests, yet their share of global deaths remained high, suggesting under-detection of infections and weaker health system capacity. For WHO, this underscores the importance of scaling up testing infrastructure in regions where limited detection capacity may hide the true scale of vulnerability.

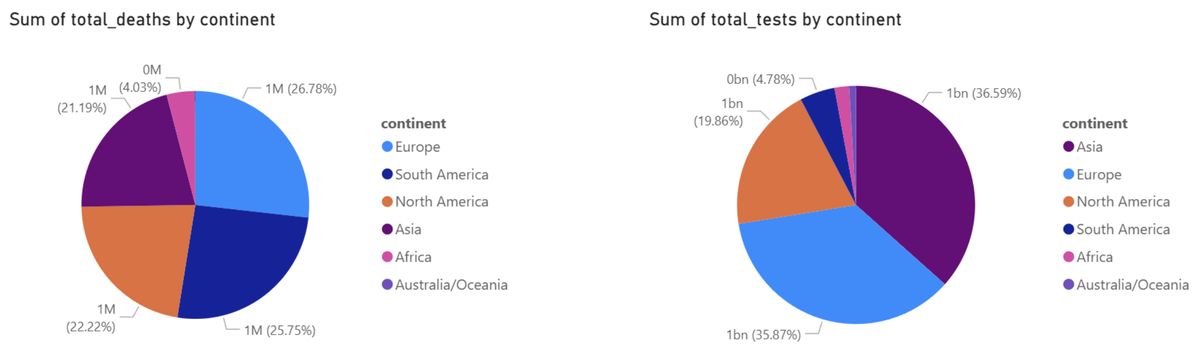


Figure 14: COVID-19 Total deaths vs Total tests across continents

3.1.1 Recommendations

The combined analysis from Excel and Power BI highlights both the scale of the COVID-19 pandemic and the structural differences that shaped outcomes across continents. The evidence suggests that WHO resource allocation should focus on strengthening health systems in regions with high death-to-case ratios, limited vaccination coverage, and weaker testing and surveillance. Based on the insights, the following recommendations are proposed:

3.1.2 Expand healthcare infrastructure in vulnerable regions

Countries in Africa and South America reported disproportionately high mortality despite lower case detection, reflecting limited hospital capacity and intensive care availability. WHO should prioritize investment in:

- Expanding hospital and ICU bed capacity.
- Improving oxygen supply chains and emergency care.
- Training and retaining frontline healthcare workers.

Strengthening these systems will reduce vulnerability to future outbreaks. This aligns with Uppal et al., (2020), who emphasize the role of high-quality health systems in reducing preventable deaths.

3.1.3 Strengthen Vaccination access and equity

Vaccination coverage was highest in Europe and North America but lagged significantly in Africa and parts of Oceania. This inequity widened gaps in mortality outcomes. WHO should:

- Increase vaccine procurement and distribution channels in low-coverage countries.

- Support regional manufacturing hubs to reduce dependency on external supply.
- Invest in public health campaigns to address vaccine hesitancy.

Global equity in vaccine access is critical to reducing mortality.

3.1.3 Scale Up Testing and Surveillance Systems

Power BI analysis showed that continents with lower testing (Africa and South America) also recorded disproportionately high deaths, suggesting under-detection. WHO should:

- Support the development of scalable, low-cost testing infrastructure.
- Standardize data reporting and monitoring systems.
- Integrate genomic surveillance to detect variants early.

This recommendation reflects findings by Wang et al., (2024) who stress the importance of robust surveillance in outbreak control.

3.1.4 Invest in Community Resilience and Recovery Capacity

Recovery rates varied widely, with North America and parts of Africa lagging. Community-based interventions, local health worker training, and improved access to essential medicines are needed to strengthen recovery capacity. WHO should also support partnerships with local organizations to enhance trust and ensure public health guidance is followed.

As Zahner (2005) notes, building resilience at the community level is key to long-term pandemic preparedness.

4.1.1 Conclusion

This report analyzed COVID-19 data using Excel and Power BI to provide the World Health Organization (WHO) with evidence-based insights for resource allocation. The Excel dashboard gave a descriptive overview of the pandemic's scale, highlighting that Europe and the Americas carried the heaviest confirmed case burdens, while South America and Africa faced higher death-to-case ratios due to weaker healthcare systems. The Power BI dashboard added a diagnostic layer, showing how vaccination inequities, limited hospital capacity, and low testing levels explained differences in outcomes across regions.

The analysis leads to four key recommendations for WHO: expand healthcare infrastructure in vulnerable regions, strengthen vaccination access, scale up testing and surveillance, and invest in community resilience. Together, these actions target the structural weaknesses revealed in the data and would help prepare low- and middle-income countries for future health crises.

At the same time, the study highlights important limitations in the dataset. Reporting inconsistencies, under-testing, and missing socioeconomic indicators mean that the true scale of the pandemic in some countries is likely underestimated. Mortality rates in under-reported regions may be higher than the data suggests. Future improvements in data collection should include broader demographic, economic, and health system variables, alongside standardized reporting protocols, to provide a more accurate foundation for decision-making.

In conclusion, the dashboards demonstrate the value of visual analytics in turning complex global health data into actionable insights. By combining descriptive and diagnostic perspectives, this report shows that case numbers alone are insufficient for understanding vulnerability. Mortality ratios, healthcare readiness, and recovery capacity offer a clearer picture of where WHO support is most urgently needed. Strengthening these systems is not only a matter of equity for the most affected regions but also a global necessity, as the next pandemic will test the preparedness of all nations.

5.1.1References

Abdullah Al Maruf, Paul, R., Imam, M. H., & Babar, Z. (2022). A SYSTEMATIC REVIEW OF THE ROLE OF SQL AND EXCEL IN DATA-DRIVEN BUSINESS DECISION-MAKING FOR ASPIRING ANALYSTS. *A SYSTEMATIC REVIEW of the ROLE of SQL and EXCEL in DATA-DRIVEN BUSINESS DECISION-MAKING for ASPIRING ANALYSTS*, *01*(01), 249–269. https://doi.org/10.63125/n142cg62

Ayoade Alakija. (2023). Leveraging lessons from the COVID-19 pandemic to strengthen low-income and middle-income country preparedness for future global health threats. *The Lancet Infectious Diseases*, *23*(8). https://doi.org/10.1016/s1473-3099(23)00279-7

Ayogeboh Epizitone, Smangele Pretty Moyane, & Israel Edem Agbehadji. (2023). A Data-Driven Paradigm for a Resilient and Sustainable Integrated Health Information Systems for

Health Care Applications. *Journal of Multidisciplinary Healthcare*, *Volume 16*, 4015–4025. https://doi.org/10.2147/jmdh.s433299

Chowdhury, R., Luhar, S., Khan, N., Choudhury, S. R., Matin, I., & Franco, O. H. (2020). Long-term strategies to control COVID-19 in low and middle-income countries: an options overview of community-based, non-pharmacological interventions. *European Journal of Epidemiology*, *35*(8), 743–748. https://doi.org/10.1007/s10654-020-00660-1

Haider, N., Osman, A. Y., Gadzekpo, A., Akipede, G. O., Asogun, D., Ansumana, R., Lessells, R. J., Khan, P., Hamid, M. M. A., Yeboah-Manu, D., Mboera, L., Shayo, E. H., Mmbaga, B. T., Urassa, M., Musoke, D., Kapata, N., Ferrand, R. A., Kapata, P.-C., Stigler, F., & Czypionka, T. (2020). Lockdown Measures in Response to COVID-19 in Nine sub-Saharan African Countries. *BMJ Global Health*, *5*(10).

Santangelo, O. E., Provenzano, S., Di Martino, G., & Ferrara, P. (2024). COVID-19 Vaccination and Public Health: Addressing Global, Regional, and Within-Country Inequalities. *Vaccines*, *12*(8), 885–885. https://doi.org/10.3390/vaccines12080885

Skolnik, A., Bhatti, A., Larson, A., & Mitrovich, R. (2021). Silent Consequences of COVID-19: Why It's Critical to Recover Routine Vaccination Rates Through Equitable Vaccine Policies and Practices. *The Annals of Family Medicine*, 2730. https://doi.org/10.1370/afm.2730

Taylor, L. (2021). Covid-19: Why Peru suffers from one of the highest excess death rates in the world. *BMJ*, n611. https://doi.org/10.1136/bmj.n611

Uppal, A., Silvestri, D. M., Siegler, M., Natsui, S., Boudourakis, L., Salway, R. J., Parikh, M., Agoritsas, K., Cho, H. J., Gulati, R., Nunez, M., Hulbanni, A., Flaherty, C., Iavicoli, L., Cineas, N., Kanter, M., Kessler, S., Rhodes, K. V., Bouton, M., & Wei, E. K. (2020). Critical

Care And Emergency Department Response At The Epicenter Of The COVID-19 Pandemic. *Health Affairs*, *39*(8), 1443–1449. https://doi.org/10.1377/hlthaff.2020.00901

Wang, S., Jiang, X., & Muhammad Bilawal Khaskheli. (2024). The Role of Technology in the Digital Economy's Sustainable Development of Hainan Free Trade Port and Genetic Testing: Cloud Computing and Digital Law. *Sustainability*, *16*(14), 6025–6025. https://doi.org/10.3390/su16146025

World Health Organisation. (2017). *Health Systems Resilience*. Www.who.int. https://www.who.int/teams/primary-health-care/health-systems-resilience

World Health Organization. (2021). *Vaccine equity*. Www.who.int; World Health Organization. https://www.who.int/campaigns/vaccine-equity

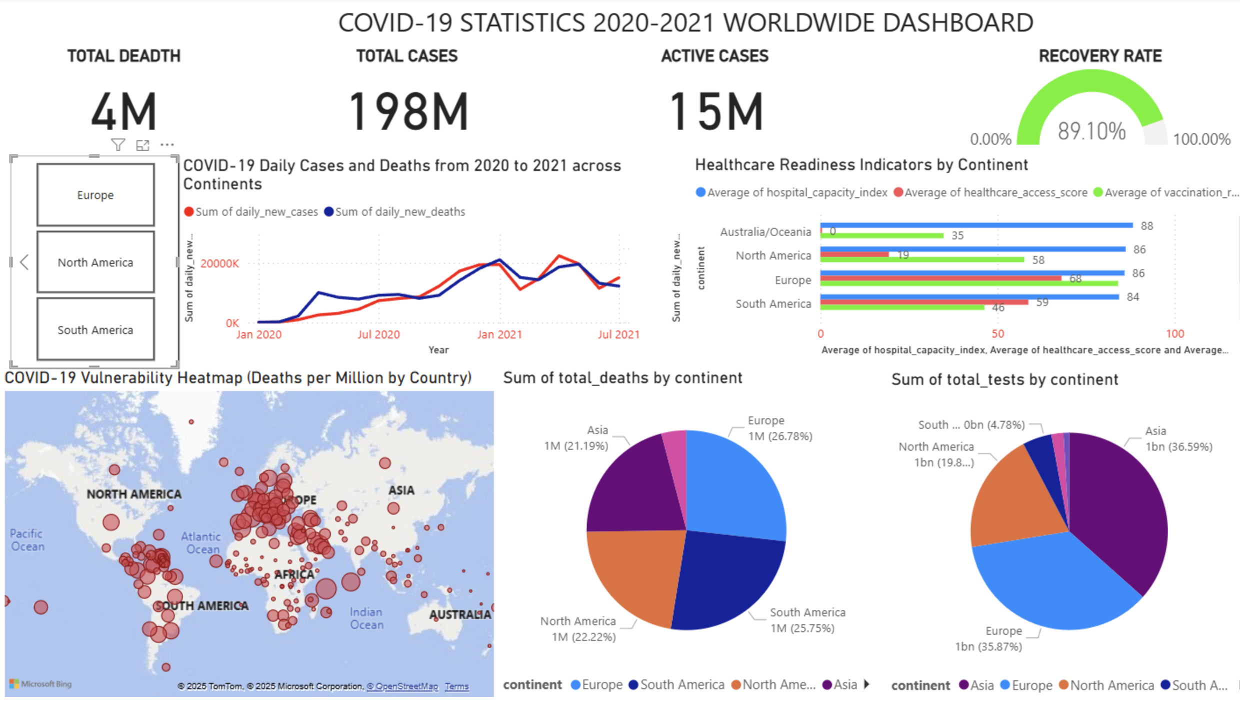
World Health Organization. (2022). *The COVID-19 pandemic and continuing challenges to global health*. Www.who.int. https://www.who.int/about/funding/invest-in-who/investment-case-2.0/challenges

Wu, Z., Lin, L., Zhang, J., Zhong, J., & Lai, D. (2025). Global burden of diphtheria, 1990–2021: a 204-country analysis of socioeconomic inequality based on SDI and DTP3 vaccination differences before and after the COVID-19 pandemic (GBD 2021). *Frontiers in Public Health*, *13*. https://doi.org/10.3389/fpubh.2025.1597076

Zahner, S. J. (2005). Local Public Health System Partnerships. *Public Health Reports*, *120*(1), 76–83. https://doi.org/10.1177/003335490512000113

6.1.1 Appendix

6.1.2Power Bi Dashboard



6.1.3 Excel Dashboard